Week 10 Summary - Big Data Architecture for Real-Time Fraud Detection

Problem Statement

A global bank faces rising fraud and anti-money laundering (AML) pressure across high-volume transaction channels. Legacy batch detection delays response, increases losses, and creates a high false-positive burden for investigators. The bank needs real-time detection, automated high-risk blocking, anomaly analysis, customer security alerts, and auditable compliance controls.

Proposed Architecture (High Level)

1) NiFi ingests and routes card, account, and customer activity streams from multiple systems.
2) Kafka provides a durable event backbone for continuous processing.
3) Spark performs streaming feature engineering, fraud scoring, and rule-based detection.
4) HBase stores low-latency customer profiles and recent risk context for real-time lookups.
5) HDFS stores historical transaction data for long-term analysis and model improvement.
6) Hive executes AML analytics and produces compliance-ready reporting datasets.
7) Solr powers investigator search, triage, and fast case retrieval.
8) YARN orchestrates cluster resources to keep Spark/Hive workloads reliable and scalable.

Benefits and Conclusion

- Faster prevention: suspicious transactions can be blocked within seconds.
- Better precision: behavior-based analytics improve detection quality and reduce false positives.
- Compliance strength: auditable data pipelines and reporting support AML obligations.
- Enterprise scale: distributed components support global transaction growth.
- Trust protection: reduced fraud losses and stronger customer confidence.

Conclusion: The architecture combines real-time intelligence and regulatory accountability, making it a strong fit for a global bank fraud and AML program.